11.2.18 Matgeo

AI25BTECH11012 - Garige Unnathi

Question

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of it.

Let ABC be the triangle ,
E and **F** be the midpoints of AB and BC respectively .

$$\mathbf{E} = \frac{A+B}{2} \tag{1}$$

$$\mathbf{F} = \frac{B+C}{2} \tag{2}$$

Solution

The line joining these two points is

$$\mathbf{EF} = \frac{C - A}{2} \tag{3}$$

The line AC is

$$\mathbf{AC} = \mathbf{C} - \mathbf{A} \tag{4}$$

Hence ,

$$\mathbf{EF} = \frac{1}{2}\mathbf{AC} \tag{5}$$

Therefore the line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of it